Exploring Employment Trends Among Nepali Students 2023-2024

1. Project Overview

This project focuses on analyzing employment trends, educational preferences, entrance examination participation, academic performance, and career interests among Nepali students using Microsoft Power BI.

The dashboard was designed to convert raw educational data into meaningful business insights through interactive KPI cards, slicers, and analytical visualizations. The project demonstrates practical applications of business intelligence concepts including data cleaning, data preprocessing, data modeling, DAX calculations, and dashboard storytelling.

The final dashboard helps understand student educational behavior, career preferences, and entrance examination trends in Nepal.

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2. Objectives

The major objectives of this dashboard are:

1. To identify the most preferred entrance examination among students.
2. To analyze educational stream distribution among Nepali students.
3. To study gender participation trends in education.
4. To analyze students attending multiple entrance examinations.
5. To evaluate student performance levels in competitive entrance exams.
6. To identify the most preferred future job roles among students.

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3. Data Source

Source Description: The dataset contains educational, entrance examination, demographic, academic, and career preference information of Nepali students.

Timeline: 2023 – 2024

Domain: Career Analytics

Data Size: 18,180 rows and 42 columns

Tools Used

- Microsoft Excel
- Microsoft Power BI
- Power Query
- DAX

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4. Data Preprocessing (Power Query)

Data cleaning and preprocessing were performed using Power Query Editor in Power BI.

Data Cleaning Highlights

- Removed fully blank rows.
- Removed invalid records with Student_ID = 0.
- Removed unnecessary null values.
- Filtered blank categories from important fields.
- Corrected numerical data types.
- Standardized categorical values.
- Cleaned performance and examination-related columns.
- Improved dataset consistency for accurate dashboard analysis.

These preprocessing steps improved data quality and dashboard reliability.

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5. Data Modeling

The project used data modeling techniques within Power BI to establish analytical calculations and relationships.

5.1 Data Modeling Activities

- Created calculated columns using DAX.
- Created KPI measures using DAX functions.
- Developed entrance examination participation analysis.
- Created performance classification columns.
- Built measures for attendee counts and GPA calculations.
- Implemented analytical calculations for multiple exam participation.

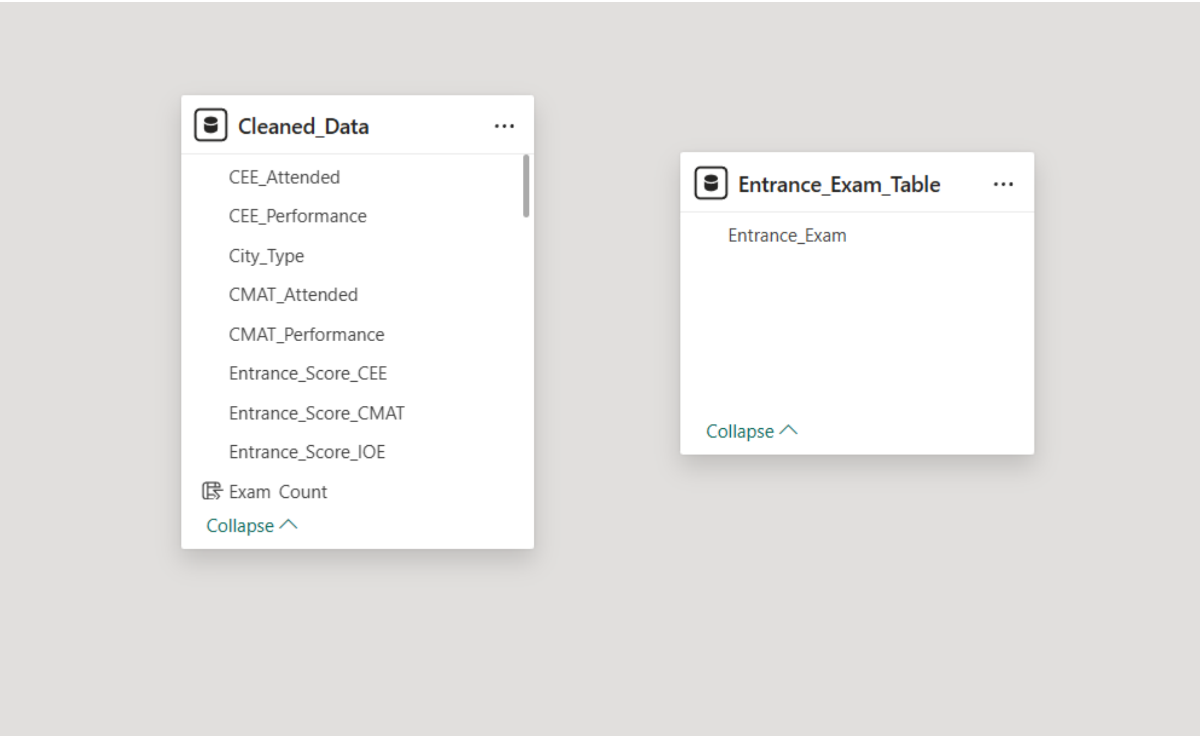
5.2 DAX Measures Created

- Total Students
- IOE Attendees
- CEE Attendees
- CMAT Attendees
- Multiple Exam Students
- Average GPA

5.3 Schema Used:

Flat schema with a primary fact table (Cleaned_Data) and one disconnected helper table (Entrance_Exam_Table) for dynamic visualization analysis.

5.4 Data Modeling Screenshot



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6. Dashboard Design

The dashboard was designed using a clean and consistent layout with interactive visuals and slicers.

6.1 KPI Cards Implemented

- Total Students
- IOE Attendees
- CEE Attendees
- CMAT Attendees
- Multiple Exam Students
- Average GPA

6.2 Slicers Used

- Gender
- Educational Stream
- Family Income

6.3 Visuals Implemented

Visual	Chart Type	Description
Most Preferred Entrance Examination	Column Chart	Shows student preference for entrance examinations
Gender Distribution Analysis	Donut Chart	Displays male and female participation trends
Educational Stream Distribution	Treemap	Shows distribution of educational backgrounds
Students Attending Multiple Entrance Exams	Column Chart	Analyzes students attempting multiple exams
IOE Performance Analysis	Funnel Chart	Displays performance-level distribution
Top Preferred Job Roles	Treemap	Shows most preferred career choices

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7. Key Insights

7.1. Entrance Examination Preference Trends

Descriptive Analytics

IOE entrance examination recorded the highest participation among students, showing strong preference toward engineering-related higher education.

Diagnostic Analytics

The increasing popularity of engineering, technology, and technical careers may have influenced higher participation in IOE examinations.

Predictive Analytics

Technical entrance examinations are expected to continue dominating student preferences in future academic years.

Prescriptive Analytics

Educational institutions can strengthen STEM-focused programs, technical workshops, and engineering preparation support.

7.2. Gender Participation Analysis

Descriptive Analytics

The dataset shows relatively balanced participation between male and female students.

Diagnostic Analytics

Improved awareness and accessibility of higher education opportunities may have contributed to balanced participation.

Predictive Analytics

Gender representation in competitive examinations and higher education is expected to remain balanced in the coming years.

Prescriptive Analytics

Institutions should continue promoting inclusive education policies and equal learning opportunities.

7.3. Educational Stream Trends

Descriptive Analytics

Science-related educational streams contain the highest number of students in the dataset.

Diagnostic Analytics

Students are increasingly attracted toward science backgrounds because of strong career opportunities in engineering, medicine, and IT sectors.

Predictive Analytics

Science and technology-based education streams are expected to remain highly preferred among students.

Prescriptive Analytics

Educational institutions can improve laboratory facilities, technical infrastructure, and STEM learning resources.

7.4. Multiple Entrance Examination Participation

Descriptive Analytics

A significant number of students attended more than one entrance examination.

Diagnostic Analytics

Students may attempt multiple examinations to maximize admission opportunities and career flexibility.

Predictive Analytics

Competition among students is expected to increase, resulting in higher multi-exam participation rates.

Prescriptive Analytics

Career counseling and educational guidance programs can help students make focused academic decisions.

7.5. Student Performance Analysis

Descriptive Analytics

Most students fall under the Average Performer category in entrance examination performance.

Diagnostic Analytics

The data indicates moderate preparation levels and varying academic readiness among students.

Predictive Analytics

Students with stronger academic backgrounds and higher GPA scores are more likely to achieve better competitive exam performance.

Prescriptive Analytics

Institutions can provide mentoring programs, mock examinations, and additional coaching support for students.

7.6. Career Preference Analysis

Descriptive Analytics

Engineering, IT, healthcare, and business-related professions are among the most preferred job roles.

Diagnostic Analytics

Students are more attracted toward careers that offer strong salary potential, job stability, and future growth opportunities.

Predictive Analytics

Technology-oriented and professional careers are expected to dominate future employment trends.

Prescriptive Analytics

Educational institutions can strengthen industry-oriented training, technical skill development, and career guidance initiatives.

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8. Analytical Insights

1. Approximately 6K+ students attended the IOE entrance examination, making it the most preferred entrance exam among students.
2. Science-related educational streams contributed the highest student share, accounting for more than 40% of the overall dataset.
3. Gender participation remained relatively balanced, with both male and female students contributing significantly to higher education participation.
4. More than 3K students attended multiple entrance examinations, indicating high academic competition and broader career exploration.

5. The Average Performer category contained the largest number of students in the IOE performance analysis, showing moderate preparation levels among candidates.
6. Engineering and IT-related professions emerged as the top preferred job roles among students.
7. The dashboard analyzed more than 18,000 student records, providing broad insights into educational and employment trends.
8. Students with higher GPA scores demonstrated better entrance examination participation and performance patterns.
9. Technical and professional careers are expected to remain dominant future employment choices among Nepali students.

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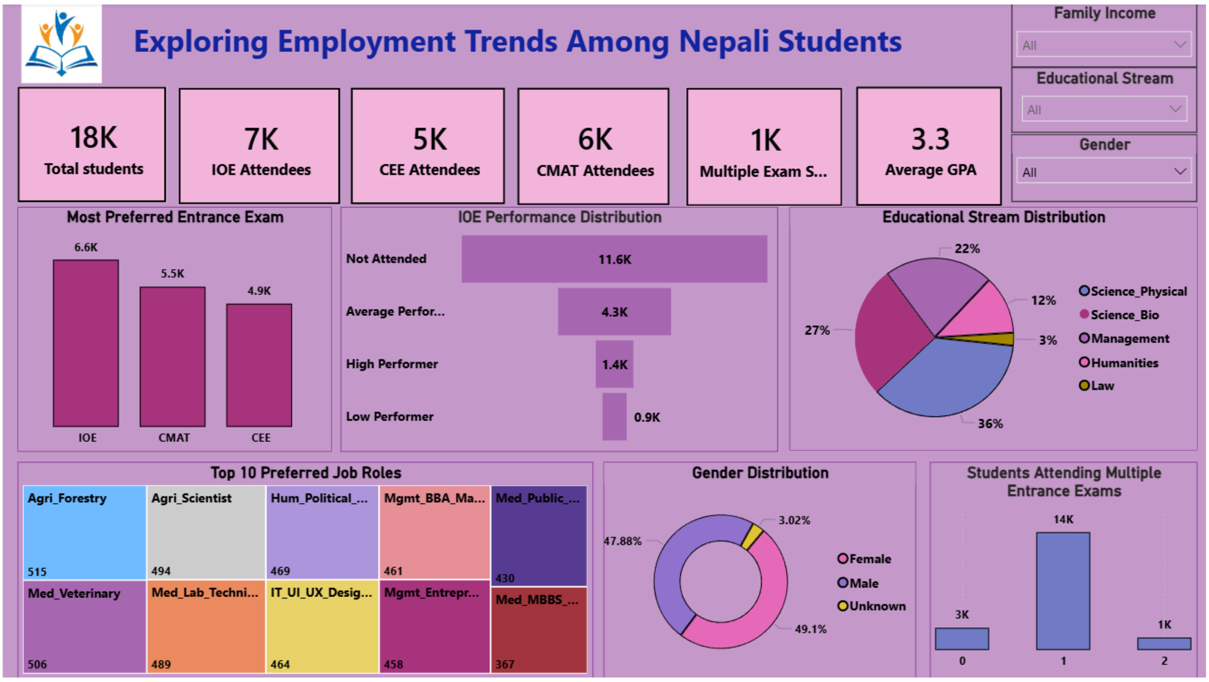
9. Dashboard Screenshot



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10. Conclusion

This project successfully analyzed educational preferences, entrance examination participation, academic performance, and career interests among Nepali students using Power BI.

The dashboard provides meaningful business insights through KPI indicators, interactive visualizations, and analytical storytelling techniques.

The project demonstrates practical implementation of:

- Data cleaning
- Data preprocessing
- Data modeling
- DAX calculations
- Dashboard development
- Business intelligence concepts

The findings can help educational institutions and students better understand educational trends and future career preferences among Nepali students.
